

Early Prediction of Movie Box Office Success Based on Wikipedia Activity Data

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Can Wikipedia Activity Predict Movie Box Office Success Before Release?

Background

- Digital platforms leave large-scale traces of public attention and behavior  Wikipedia, Twitter.etc
 - Monitor and predict how society reacts to products or events
 - stock markets, disease spread, and earthquakes
- Online collective activity can serve as a **predictive signal** for offline outcomes.
 - Whether similar signals can be used for **movie box office prediction**

Commercial value: *Early revenue prediction can guide marketing and release strategy*

Original Paper Purpose

Prior movie prediction methods had **limitations:**

- **Neural network:** 36.9% classification accuracy
- **Text regression:** $R^2 \approx 0.671$
- **Twitter model:** strong but small sample and near-release prediction

remaining gap: *Can we predict movie revenue **earlier**, with a **larger sample**, using **simple online activity measures**?*

Research Question

Can online activity on Wikipedia before a movie is released be used to predict how successful that movie will be at the box office?

Hypothesis: movies with more Wikipedia views, edits, editors, and collaborative activity before release will earn higher opening weekend revenue.

Replicate the paper's Wikipedia-based prediction model for opening weekend box office revenue

Data

Original data package:

- a **movie-level file** with financial and release information
- a folder of **Wikipedia predictor** files with **time-varying activity** measures for each movies

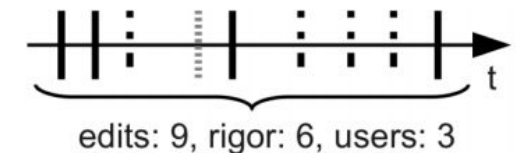
- ❑ **Dataset:** 312 movie-level instances, 5 features, no missing value
 - **Data matching:** Matched 535 Box Office Mojo movies with Wikipedia records
Final sample: 312 movies with both financial and Wikipedia activity data
 - **Source:** *Early Prediction of Movie Box Office Success Based on Wikipedia Activity Big Data*
 - **Outcome:** opening weekend revenue
 - **Predictors:** V,U,R,E,T
 - **Data Preprocessing:** renamed columns, converted variable types, and checked missing values

Variable Definitions

- **V:** *Wikipedia page views*
- **U:** *Number of human users/editors*
- **E:** *Number of human edits*
- **R:** *Collaborative rigor*
- **T:** *Number of theaters*

wikipedia activity measures

opening weekend theater count



Construct predictors at $t = -30$ using only pre-release Wikipedia activity

Model

- **Movie-time setup:**
 - $t = 0$: movie release date
 - $t < 0$: days before release
 - $t > 0$: days after release
- **Prediction cutoff:**
 - $t = -30$ Original paper's one month ahead prediction setting
 - For each movie, we use only Wikipedia records with **movie_time** ≤ -30
 - Predictor values are **accumulated activity measures** from article inception up to $t = -30$
- **Construction steps:**
 - Read each movie's Wikipedia predictor file
 - Keep rows with **movie_time** ≤ -30
 - Select the **latest cumulative values** of **V**, **U**, **R**, and **E** before the cutoff
 - Merge these predictors with **revenue** and **theaters** by **movie_id**
 - Construct a final movie-level dataset with one row per movie
 - Final dataset: 312 movie-level observations with 5 predictors.

	movie_id <int>	title <chr>	revenue <dbl>	theaters <dbl>	V <dbl>	U <dbl>	R <dbl>	E <dbl>
1	1	127 Hours	264851	4	326587	77	89	117
2	2	2012 Time for Change	9009	1	0	0	0	0
3	3	44 Inch Chest	4185	1	83258	22	22	32
4	4	45365	107	1	274	0	0	0
5	5	71 Into the Fire	77161	20	3757	4	4	6
6	6	8 The Mormon Proposition	42566	16	2058	7	9	29

Avoids data leakage by excluding release-day, opening-weekend, and post-release information.

Fit the original linear regression model and evaluate with 10-fold cross-validation

Model setup

Model

- **Model Type:** *raw multivariate linear regression*
 - Predicts opening weekend box office revenue using different predictor sets
- **Outcome Variable:** *opening weekend revenue Y*
 - Y is the first weekend box office revenue in USD
same outcome used in the original paper
- **Main Full Model:** {V,U,R,E,T}
 - V,U,R,E: Wikipedia activity measures
 - T: number of theaters screening the movie on opening weekend

Predictor-set Comparison

- *We compare the same types of predictor sets as the original paper:*
 - {T}: theater-only benchmark
 - {V}: page-view-only model
 - {V,T}: page views + theaters
 - {U,T}: users/editors + theaters
 - {V,U,R,E}: Wikipedia-only activity model
 - {V,U,R,E,T}: full Wikipedia + theater model
- **Purpose of comparison:**
 - Test whether Wikipedia activity adds value beyond theater count
 - Identify whether the full predictor set reproduces the paper's best-performing model

10-fold cross-validation *following the original paper*

- Randomly split the 312 movies into 10 folds
- Train the model on 9 folds
- Test the model on the remaining fold
- Repeat until each fold has been used once as the test fold.

Evaluation metric: *cross-validated $R^2(t)$*

- $R^2(t)$ measures how much variation in revenue is explained by the model at time t
- Higher $R^2(t)$ means stronger prediction performance

Main prediction time: $t = -30$

- Tests one-month-ahead prediction before release
- The original paper reports about $R^2 = 0.77$ for the full model around this time

Evaluation

Replication: Results 1

The replicated dataset preserves the main distribution patterns reported in the paper

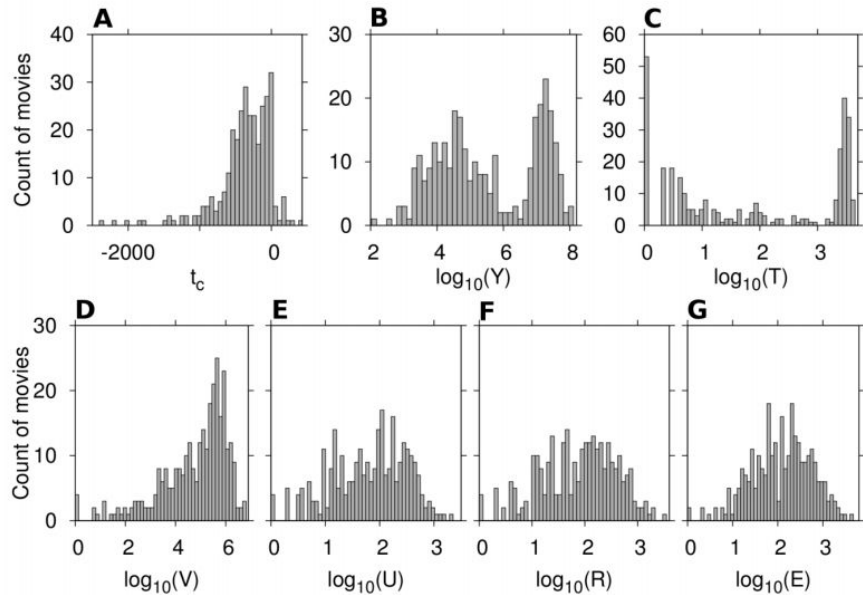
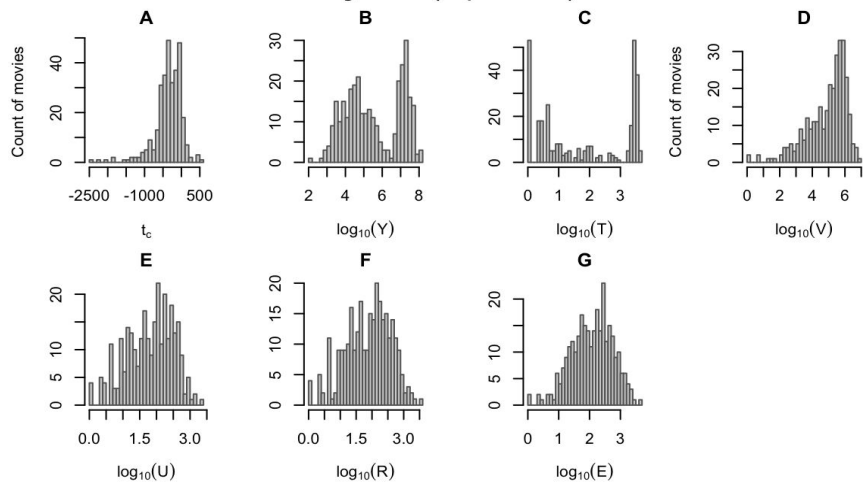


Figure 1 (replicated)



Original paper

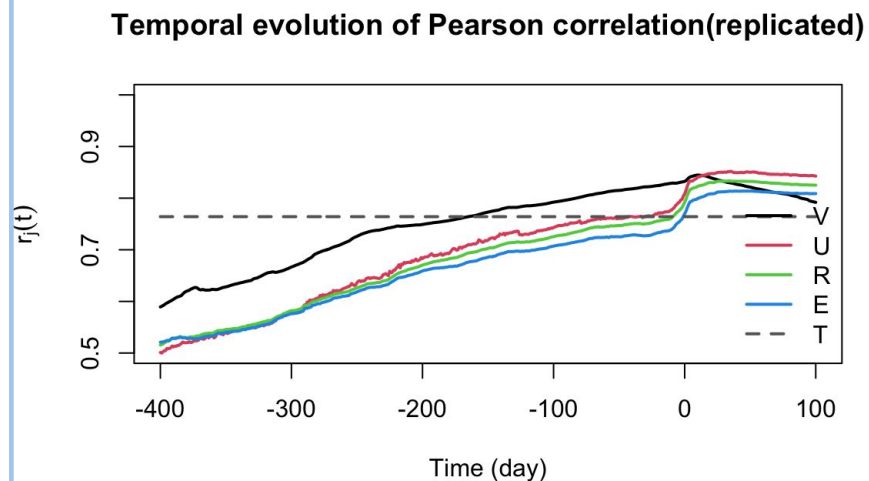
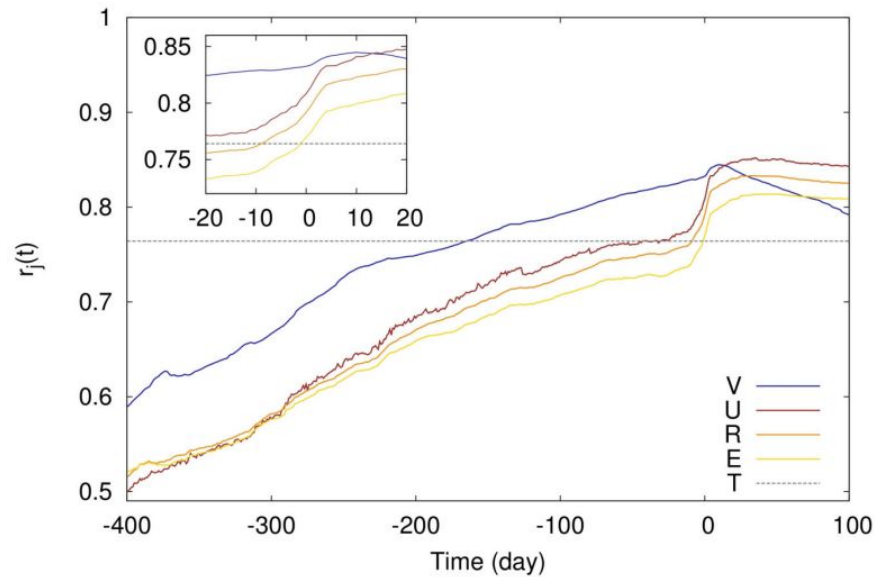
- Examines the final sample of **312 matched movies**
- Summarizes the sample using revenue Y , theaters T , and accumulated Wikipedia activity V, U, E, R up to $t=7$.
- Panel A shows article creation time t_c on the original scale
- Panels B–G show log-transformed distributions:
 - $\log_{10}(Y)$: opening weekend revenue
 - $\log_{10}(T)$: number of theaters
 - $\log_{10}(V), \log_{10}(U), \log_{10}(R), \log_{10}(E)$: Wikipedia activity measures
- Opening weekend revenue has a **bimodal distribution**.
- Wikipedia coverage is **not limited to financially successful movies**.
- V, U, R, E are **rich but skewed**: substantial activity exists, but attention is unevenly distributed across movies.

Replication

- Replicated Figure 1 applies the same log transformation for Y, T, V, U, R, E
- The replicated distributions are broadly consistent with the original paper

Replication: Results 2

Wikipedia predictors become more correlated with revenue as release approaches



What Pearson correlation measures

- $r_j(t) = \text{cor}(x_j(t), Y)$
- Whether movies with higher predictor values also tend to earn higher opening weekend revenue
- Higher $r_j(t)$ means stronger predictive signal at time t

Original paper

- **Intuition:** movies with higher Wikipedia activity tend to have higher opening weekend revenue
- Correlations increase as release approaches
- V has the highest pre-release correlation with revenue

Replication

- Our replicated Figure 2 shows the same upward trend.
- At $t=-30$, V has the strongest correlation: $r=0.822$

Pearson correlations at $t = -30$:

V	U	R	E	theaters
0.8218844	0.7643955	0.7505196	0.7271445	0.7641281

Replication: Results 3

The full model {V,U,R,E,T} reproduces the original paper's main R^2 result

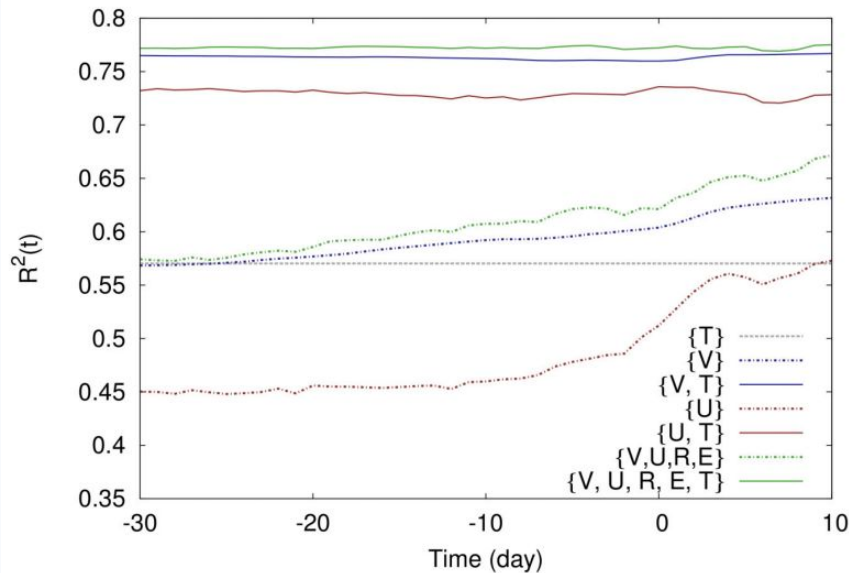
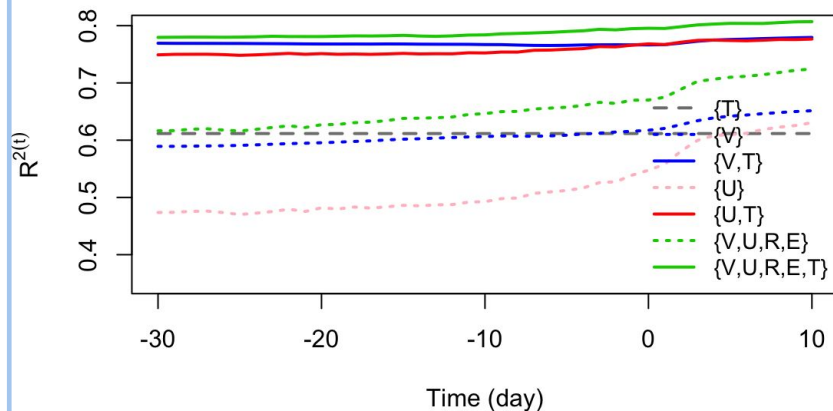


Figure 3: CV R^2 by Predictor Set(replicated)



Original paper

- The paper compares multivariate linear regression models using different predictor sets.
- Model performance is measured by $R^2(t)$ with **10-fold cross-validation**.
- The theater-only model {T} is used as the market benchmark.
- {V} alone performs roughly similarly to {T}.
- Adding Wikipedia activity to theaters improves prediction.
- The full model {V,U,R,E,T} performs best and reaches about **$R^2 = 0.77$** around one month before release.

Replication

- Our replicated Figure 3 follows the same ranking pattern.
- The full model {V,U,R,E,T} is the best-performing model.
- At $t=-30$, our full model reaches: $R^2 = 0.78$
- {V,T} also performs strongly with $R^2 = 0.769$, showing that page views add predictive value beyond theater count.

R^2 results at $t = -30$:

	features	mean_r2	sd_r2
6	V, U, R, E, theaters	0.7795215	0.11293730
3	V, theaters	0.7693135	0.12750364
4	U, theaters	0.7490607	0.09721682
5	V, U, R, E	0.6164640	0.23190472
1	theaters	0.6115873	0.13301035
2	V	0.5891894	0.25275418

Replication: Results 4

Wikipedia provides useful prediction earlier than the Twitter-based benchmark

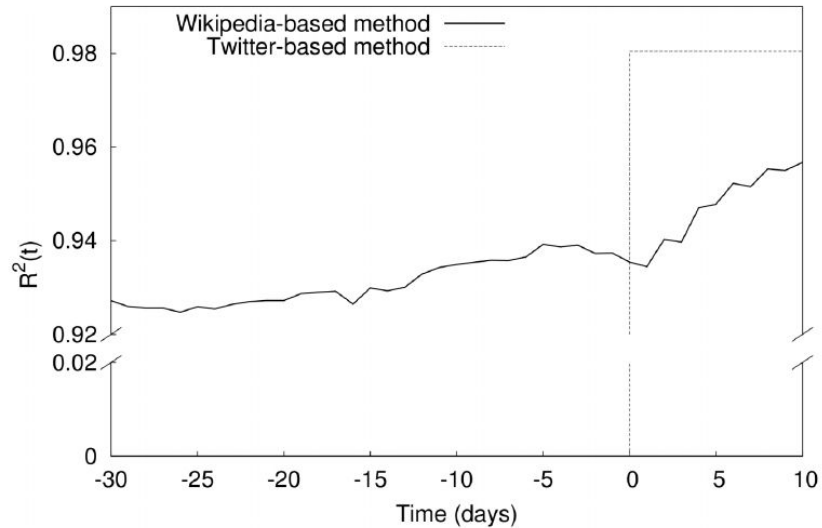
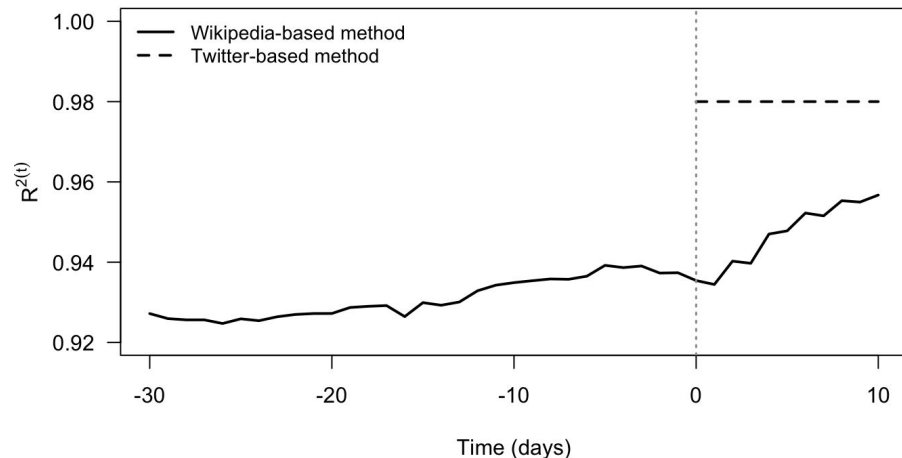


Figure 4 (replicated)



Original paper

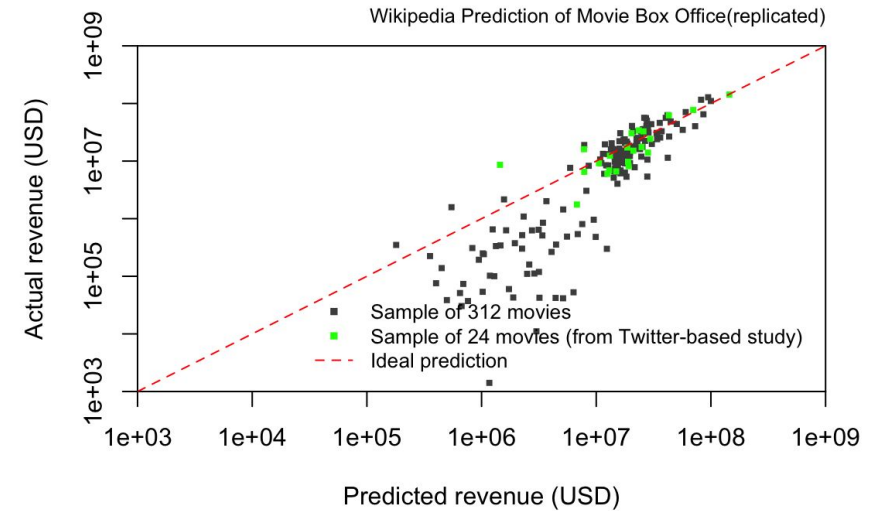
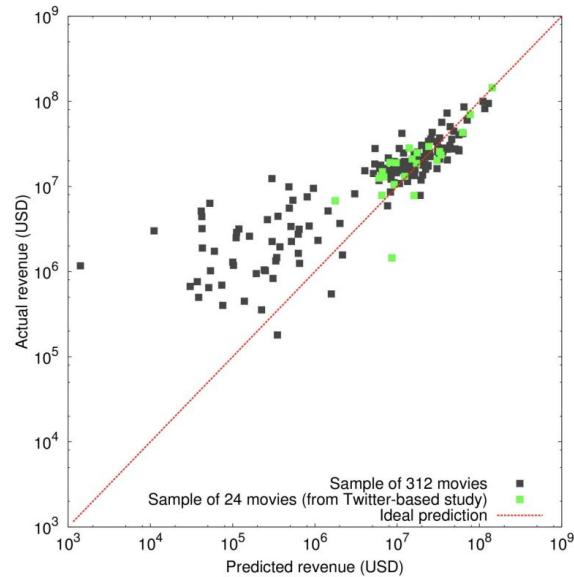
- The paper compares the Wikipedia-based method with the Twitter-based model from Asur and Huberman.
- The comparison uses the same smaller sample of **24 movies**.
- The Twitter model reaches about **$R^2=0.98$** on the night of release.
- The original paper reports that the Twitter model reaches $R^2 = 0.98$ at release, while the Wikipedia model remains above roughly $R^2 = 0.925$ around one month before release.

Replication

- Our replicated Figure 4 follows the same pattern.
- The Twitter line is higher at release.
- The Wikipedia line is slightly lower but already strong before release.
- Around $t=-30$, the Wikipedia model is already above roughly $R^2=0.925$.
- Our replicated Figure 4 preserves this pattern.

Replication: Results 5

Prediction errors are larger for low-revenue movies, but the direction of bias differs in our replication



Original paper

- Compares predicted vs. actual revenue at $t = -30$ to $t = 30$.
- High-revenue movies are predicted more accurately.
- Low-revenue movies deviate more from the ideal prediction line.
- Reason: popular movies have stronger Wikipedia activity signals.

Replication

- Same overall limitation appears: low-revenue movies are harder to predict.
- **Difference:** original plot suggests low-revenue movies are often **overpredicted**, while our replicated plot shows more **underprediction**.
- **Checked:** x-axis = predicted revenue, y-axis = actual revenue, so this is not an axis-reversal issue.

Uneven prediction accuracy in the original model motivates a more stable extension

Limitation from original paper:

- More precise predictions for successful movies
- Larger deviations for less successful movies
- Lower-revenue movies often have weaker Wikipedia signals
- Model performance depends on the amount of available public attention

Why It Matters

- Raw revenue is highly skewed
- Blockbusters may dominate OLS fit
- Prediction stability varies by revenue level
- Reliable early prediction supports marketing and release decisions

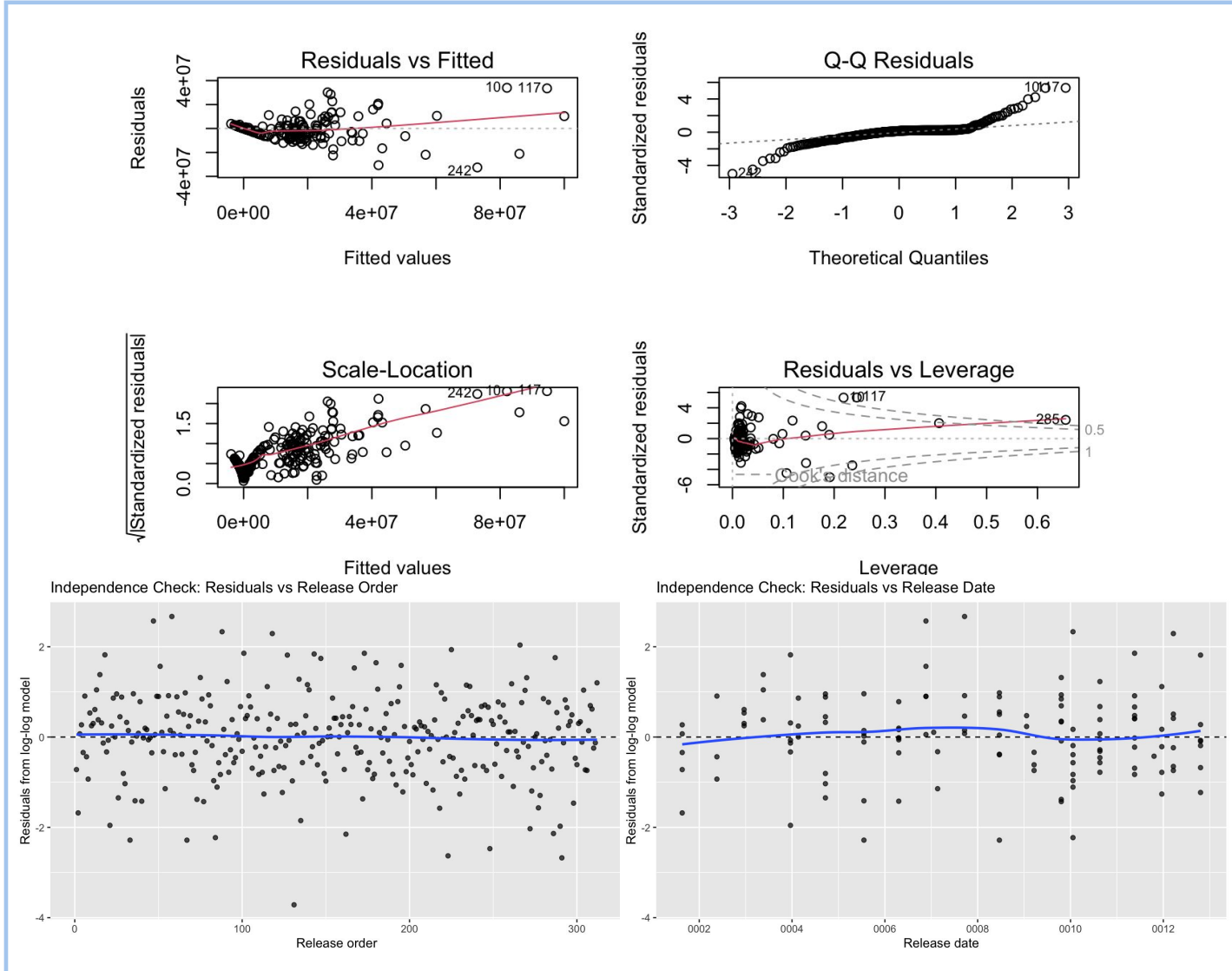


Extension Direction

- Improve model structure
- Reduce blockbuster influence
- Use only pre-release predictors
- Compare log-OLS and Gamma GLM

Given that the original model predicts successful movies more accurately than less successful movies, can we improve model stability and reduce the influence of extreme revenue values in early box office prediction?

Why Raw OLS May Be the Problem



4 Linear Assumptions

Linearity *Residuals vs Fitted*

- Red smooth line appears curved
- Revenue may not increase by a fixed dollar amount as predictors increase

Homoskedasticity *Scale-Location*

- Red smooth line increases with fitted values
- Residual spread is larger for high-revenue movies

Normality *Q-Q Residuals*

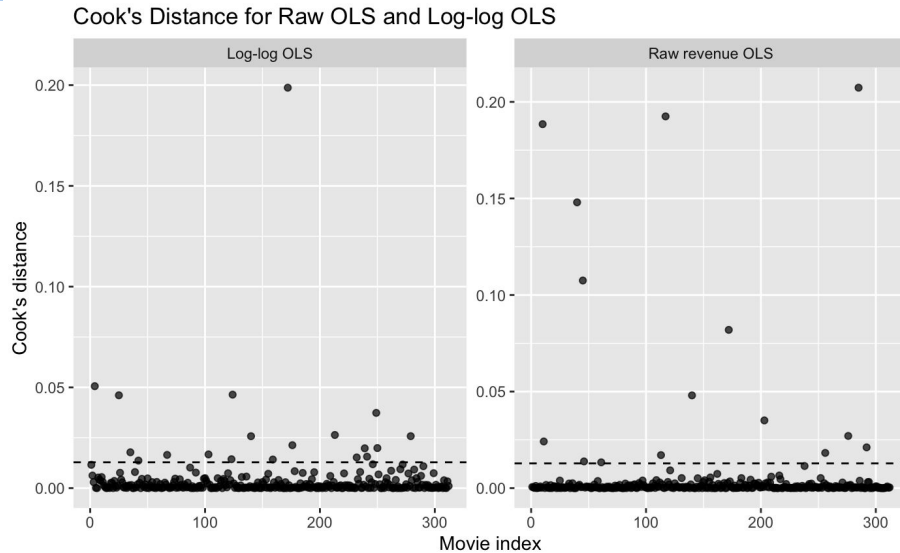
- Points deviate strongly in the tails
- Consistent with highly right-skewed revenue

Independence *Residuals vs Release Date / Order*

- No strong visual evidence of dependence

Extension Diagnostic: Influential Observations

Cook's Distance improves under log-log OLS, but one observation remains highly influential



Main Finding:

- Compared with raw OLS, **log-log OLS shows much smaller Cook's Distance values overall.**
- the log transformation **reduces the influence of extreme revenue observations.**
- **one movie still stands out highly above the threshold.**

Because this movie does not appear to be a clear data error, we **keep it in the main analysis**. We use this removal exercise as a **robustness check**, showing that the log-log model is more stable overall, although some coefficients remain sensitive.

Most influential case

- The most influential movie in the log-log model is **“Our Family Wedding”**.
 - relatively high theater count (**1605 theaters**)
 - moderate revenue (**\$7.63M**)
 - unusually weak Wikipedia activity signals
- This unusual combination gives it **high leverage and high influence**.

Coefficient sensitivity check

- Removed the most influential movie and refit the log-log model.
- Most coefficients stayed directionally similar, but some changed noticeably:
 - **log_theaters** changes only slightly:
 - **log_V** also remains very stable:
 - Larger changes appear in some editing-related variables:
 - **log_R**: 0.7521→1.02035.6%); **log_E**: -0.4269→-0.6258(-46.6%)
 - **Be careful with log_U**:
 - percent change looks large (-73.2%)
 - but the absolute change is relatively small (-0.043)

	term	full_model	without_top	absolute_change	percent_change
(Intercept)	(Intercept)	7.5073	7.5469	0.0397	0.5285
log_V	log_V	0.0422	0.0410	-0.0012	-2.9587
log_U	log_U	-0.0587	-0.1017	-0.0430	-73.1639
log_R	log_R	0.7521	1.0201	0.2680	35.6312
log_E	log_E	-0.4269	-0.6258	-0.1989	-46.5928
log_theaters	log_theaters	0.9290	0.9170	-0.0120	-1.2955

How We Improve the Model

1. Log Transformation

Purpose: *adjust the scale of skewed revenue and activity variables*

- Reduces right-skewness in revenue
- Stabilizes residual variance
- Compresses extreme blockbuster values
- Makes interpretation closer to relative change
- Reduces the influence of highly skewed Wikipedia activity

2. Gamma GLM with Log Link

Purpose: *use a model family better suited for skewed positive revenue*

- Revenue is positive and continuous
- Revenue is highly right-skewed
- Variance likely increases with the mean
- Log link keeps predicted revenue positive
- Directly connects to the GLM framework

Three Model Family:

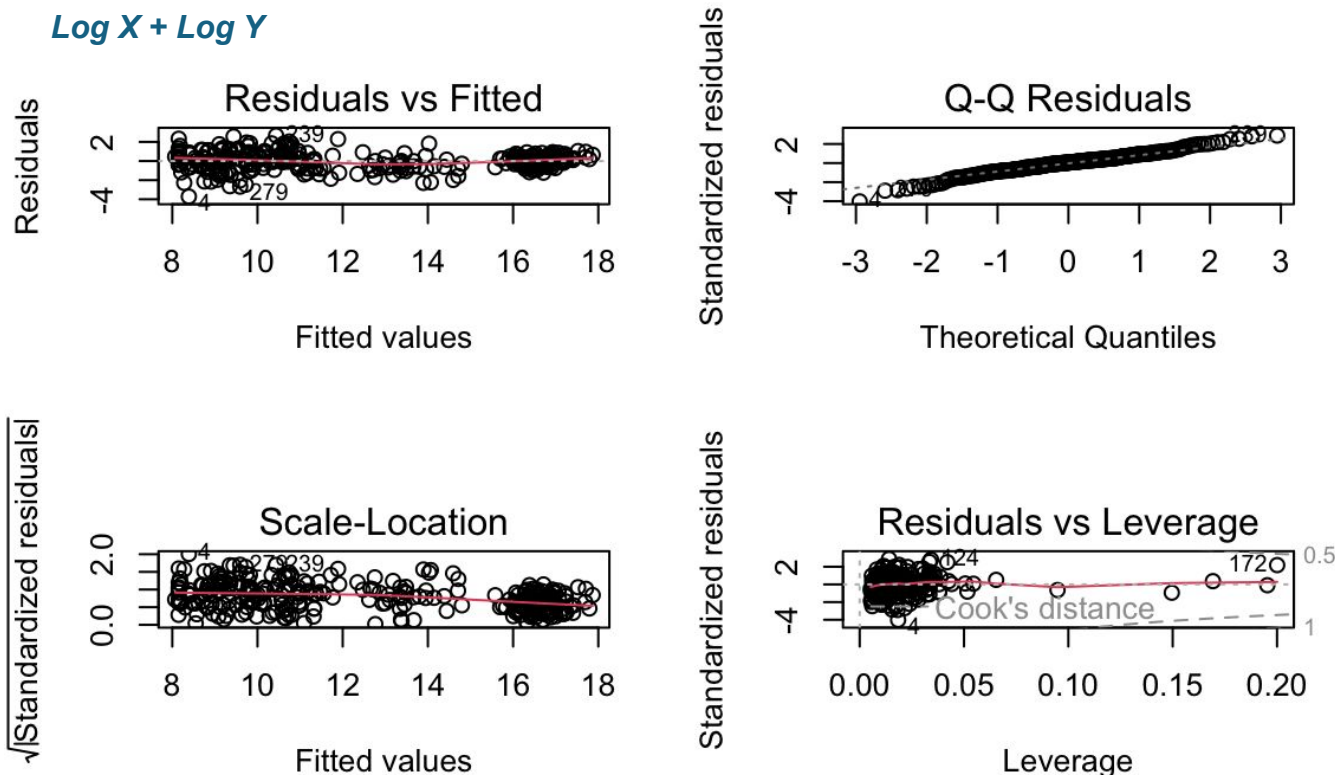
1. **Original raw revenue OLS**
2. **Log Transformation $\text{Log}X + \text{Log}Y$**
3. **Gamma GLM with Log Link**

Extension Method 1: Log-log Transformation

Both revenue and Wikipedia activity are highly skewed, so we use a log-log structure

Log X	revenue~log(1+X)
Log Y	log(revenue)~X
Log X + Log Y	log(revenue)~log(1+X)

Log X + Log Y



Why log revenue?

- Raw revenue is highly right-skewed
- Raw OLS diagnostics show scale problems
- Log revenue reduces blockbuster dominance

Why log predictors?

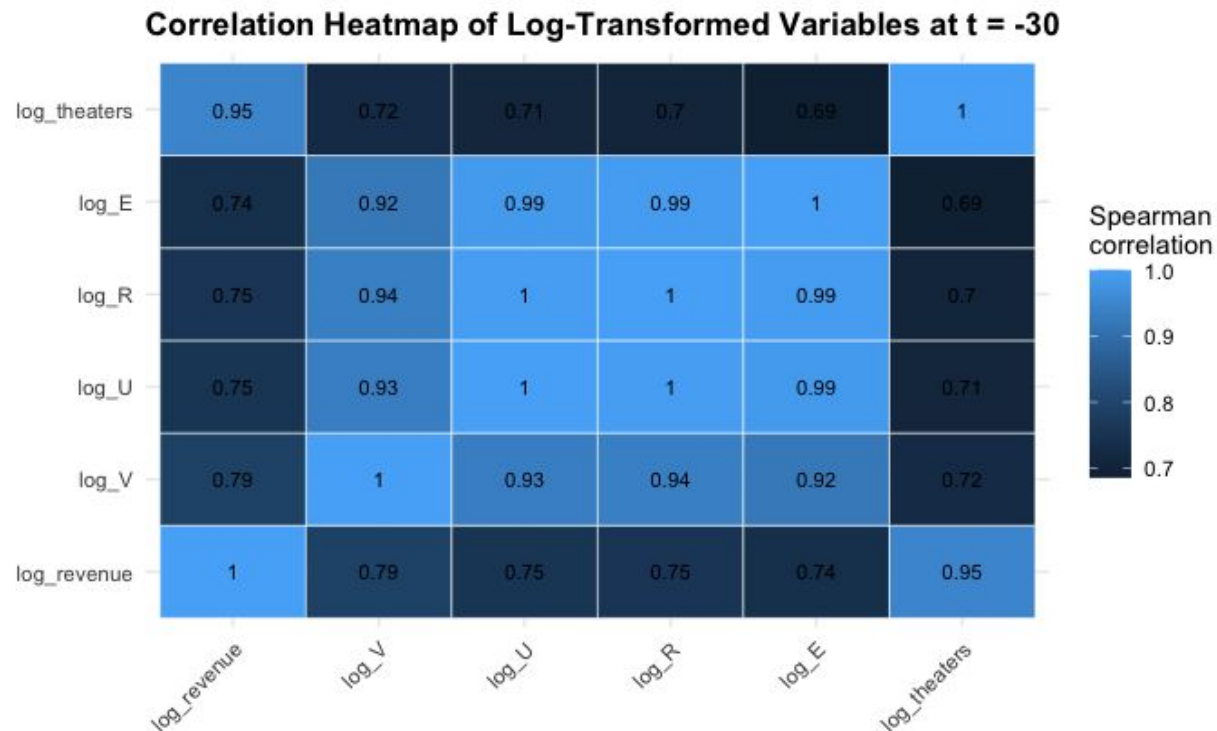
- V,U,R,E,T are also highly skewed
- $\log_{1p}(X)$ compresses extreme activity values
- Better captures diminishing returns

Use a log-log structure to address skewness on both sides.

- Residuals are more centered around zero
- Scale-Location plot becomes flatter
- Q-Q plot is closer to the reference line, though tail deviations remain
- Leverage pattern is less severe than raw OLS
- Log-log model better handles skewed revenue and extreme blockbuster values

Correlation Check Before Model Selection

Check possible overlap among Wikipedia activity variables



- ***log_theaters*** has the strongest correlation with ***log_revenue***: **0.95**
- Wikipedia activity variables are also strongly correlated with ***log_revenue***
 - ***log_V*** is the strongest Wikipedia predictor: **0.79**
- ***log_U*, *log_R*, and *log_E*** are almost **perfectly correlated**
 - Editing-related variables (U, R, E) may contain highly **overlapping information**



Reduce model

Pre-specified Log-OLS Candidate Models

Model	Predictor	Why
A: Wikipedia-only full model $\log(\text{revenue}) \sim \log1p(V) + \log1p(U) + \log1p(R) + \log1p(E)$	V,U,R,E	Heatmap: Wikipedia activity variables are all correlated with log(revenue) - Tests whether V,U,R,E can predict revenue without theater count - Comparison against models that include market distribution information
B: Wikipedia + theaters full model $\log(\text{revenue}) \sim \log1p(V) + \log1p(U) + \log1p(R) + \log1p(E) + \log1p(\text{theaters})$	V,U,R,E,T	- Closest log-log extension of the original paper's strongest predictor set - Keeps full set: V,U,R,E,TV, U, R, E, TV,U,R,E,T - Applies log transformation to address raw OLS diagnostic issues
C: Reduced log-log model $\log(\text{revenue}) \sim \log1p(V) + \log1p(U) + \log1p(E) + \log1p(\text{theaters})$	V,U,E,T	Heatmap: U,R, and E are highly correlated - R is a derived editing measure and is highly correlated with U and E - Tests whether the full model may contain overlapping editing information
D: Simple benchmark $\log(\text{revenue}) \sim \log1p(V) + \log1p(\text{theaters})$	V,T	Heatmap: V is the strongest Wikipedia predictor, and T has the strongest correlation with revenue - Tests whether the two strongest signals are sufficient - Helps evaluate whether editing variables add extra predictive value

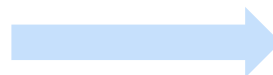
Extension Model Selection

Model Selection Within Log-OLS: Use Model B as the main model and Model C as a robustness check

model	n_folds	mean_Log_RMSE	se_Log_RMSE	mean_Log_MAE	mean_RMSLE	mean_MAE	mean_RMSE	num_predictors	AIC	BIC	delta_BIC	adj_R2
Log Model C: Reduced no rigor	10	0.924	0.065	0.708	0.924	3618876	9094357	4	850.578	873.036	0.000	0.927
Log Model B: Wikipedia + theaters	10	0.924	0.063	0.706	0.924	3498464	8727893	5	849.723	875.924	2.888	0.927
Log Model D: Simple benchmark	10	0.944	0.071	0.734	0.944	4040090	10569027	2	866.137	881.109	8.073	0.922
Log Model A: Wikipedia only	10	2.367	0.099	1.959	2.366	6815018	13116023	4	1424.776	1447.234	574.198	0.538

Selection Criteria

Goal is **prediction**, therefore prioritize out-of-sample error over in-sample fit.



Primary:

- CV MAE: typical error
- CV RMSE: penalizes large mistakes

Secondary: AIC / BIC

- Same outcome, model family, and dataset
- Only within log-OLS candidate models

Choosing Between Model B and Model C

Model B: Wikipedia + theaters full model

Main Predictive Model

- Best CV MAE and CV RMSE
- Lowest **AIC**
- Best choice when prediction accuracy is the priority

Model C: Reduced no-rigor model

Simpler Robustness Alternative

- Performs very similarly to Model B in CV
 - Removing R does not substantially reduce performance
- Lowest **BIC**
 - R overlapping with other editing-related variables

- Unstable Coefficients
- Harder interpretation
- Not generalize to other datasets

Why not Model A or D?

Performs much worse than Models B and C in CV MAE and CV RMSE

model A has highest AIC and BIC

Suggest:

- Wikipedia activity alone is not enough without theater count T
- V and T are useful, but additional Wikipedia editing variables still add predictive value

Extension Method 2: Gamma GLM with Log Link

Gamma GLM provides an alternative model family for positive, skewed revenue

Raw OLS has clear diagnostic problems

log-log OLS improves these issues, but is still linear regression

- Residuals more centered and variance is more stable
- But Q-Q plot still shows **tail deviations**



How Gamma GLM fits our problem:

- Opening weekend revenue is **positive and continuous**
- Revenue is **highly right-skewed**
- Larger expected revenue can have larger variance
- Gamma distribution is suitable for positive, skewed outcomes



Why log link?

- Keeps predicted revenue **strictly positive**
- Allows multiplicative or percentage effects
- More realistic than fixed dollar changes for box office revenue

$$Y_i \mid X_i \sim \text{Gamma}(\mu_i, \phi)$$

$$\log(\mu_i) = \beta_0 + \beta_1 \log(1 + V_i) + \dots$$

$$\text{where } \mu_i = E(Y_i \mid X_i)$$

Model setup

- Raw revenue is modeled directly
- Log link transforms expected revenue, not **Y** itself
- $\mu_i = \exp(X_i \beta)$ → Predicted revenue is positive
- $\log(1+X)$ keeps predictor scaling consistent with Log-OLS

Use Model B and Model C for Gamma GLM

Keeps **predictor scale** consistent with Log-OLS

- *Allows a fair comparison*

Model B: main predictive structure from Log-OLS selection

Model C: reduced robustness structure with R removed

Separates **model family improvement** from **predictor set choice**

Gamma Model Selection

Select Gamma Model B using AIC/BIC and 10-fold cross-validation

Model	Predictor	Why
B: Wikipedia + theaters full model $revenue \sim \log1p(V) + \log1p(U) + \log1p(R) + \log1p(E) + \log1p(theaters)$	V,U,R,E,T	- Closest Gamma version of the original full predictor set - Keeps full set: V,U,R,E,T
C: Reduced log-log model $revenue \sim \log1p(V) + \log1p(U) + \log1p(E) + \log1p(theaters)$	V,U,E,T	Tests whether removing highly correlated R gives a simpler, more stable model

model	n_folds	mean_Log_RMSE	se_Log_RMSE	mean_Log_MAE	mean_RMSLE	mean_MAE	mean_RMSE	total_bad_predictions	all_converged	num_predictors	AIC	BIC	delta_BIC
Gamma Model B: Full Wikipedia + theaters	10	1.035	0.054	0.778	1.035	3448400	7942355	0	TRUE	5	8811.862	8838.063	0.000
Gamma Model C: Reduced no rigor	10	1.038	0.053	0.786	1.037	3459187	8308900	0	TRUE	4	8818.260	8840.718	2.655

Selection Criteria:

- AIC / BIC**
 - lower values indicate better fit
- Convergence**
 - Checks whether the Gamma GLM fit is stable.
- 10-fold cross-validation**
 - metrics: Log RMSE, RMSLE, MAE, RMSE

Interpretation

- Both Gamma models converged and are valid candidates.
- Gamma B has slightly better fit and prediction accuracy:**
 - lower AIC and BIC
 - slightly lower CV RMSLE
 - lower CV MAE and RMSE
- Overall,, Gamma C does **not** improve AIC/BIC or CV performance enough to justify replacing B.

Log-log OLS model is selected as the most balanced model

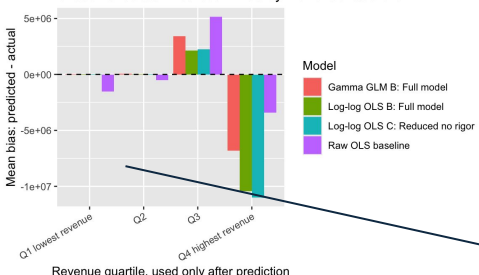
Candidate models

- **Raw OLS baseline**
 - original raw-revenue scale model
- **Log-log OLS B: Full model**
 - uses log-transformed revenue and all predictors
- **Log-log OLS C: Reduced no-rigor model**
 - uses log-transformed revenue and all predictors without R
- **Gamma GLM B: Full model**
 - models positive skewed revenue directly using Gamma response

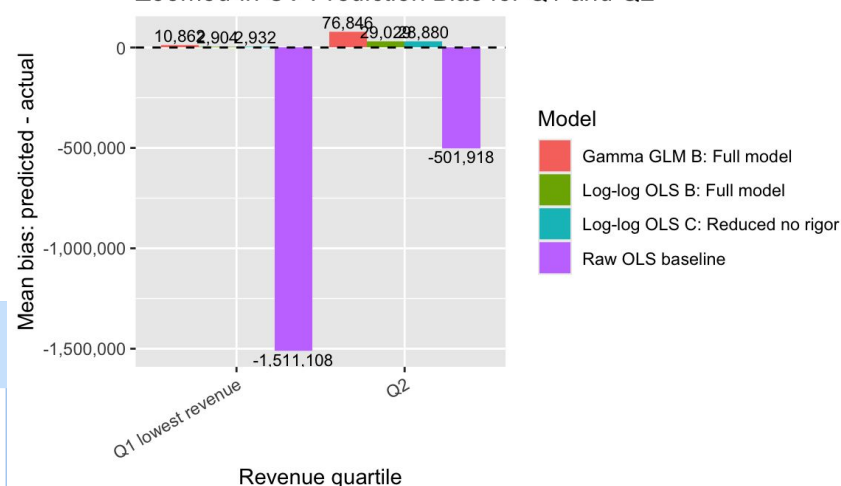
Selection Criteria

- **Overall prediction error**
 - lower RMSE is better (Root Mean Squared Log Error)
- **Performance across revenue quartiles**
 - especially Q1 and Q2
- **Bias direction and magnitude**
 - mean bias = predicted - actual
- **Model stability and interpretability**

Cross-validated Prediction Bias by Revenue Quartile



Zoomed-in CV Prediction Bias for Q1 and Q2

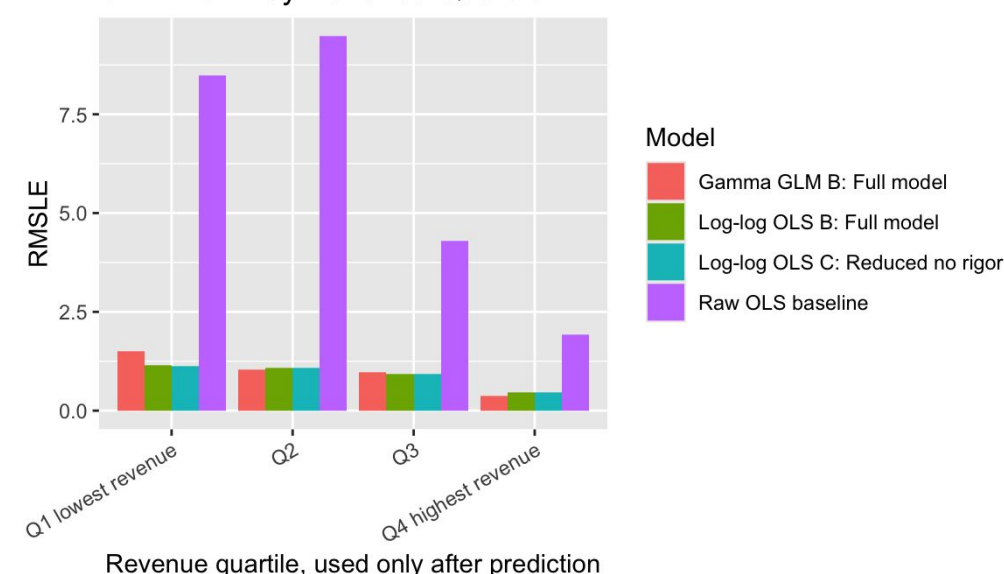


Key Findings for Bias Plot

- Raw OLS strongly underpredicts low-revenue movies.
- Gamma GLM reduces the bias substantially.
- Log-log OLS B/C are closest to zero.

log-log B has slightly better predictive accuracy, but log-log C is more simple!

CV RMSLE by Revenue Quartile



Key Findings for RMSE Plot

- Raw OLS has the largest errors across all quartiles.
- Gamma GLM improves strongly over raw OLS.
- Log-log OLS B and C are the most stable across revenue groups.

What We Would Improve in Future Work

Strength

- Uses only pre-release predictors to prevent data leakage
- Addresses skewness, blockbuster influence, and model stability
- Applies interpretable models with robustness and sensitivity checks

Limitation

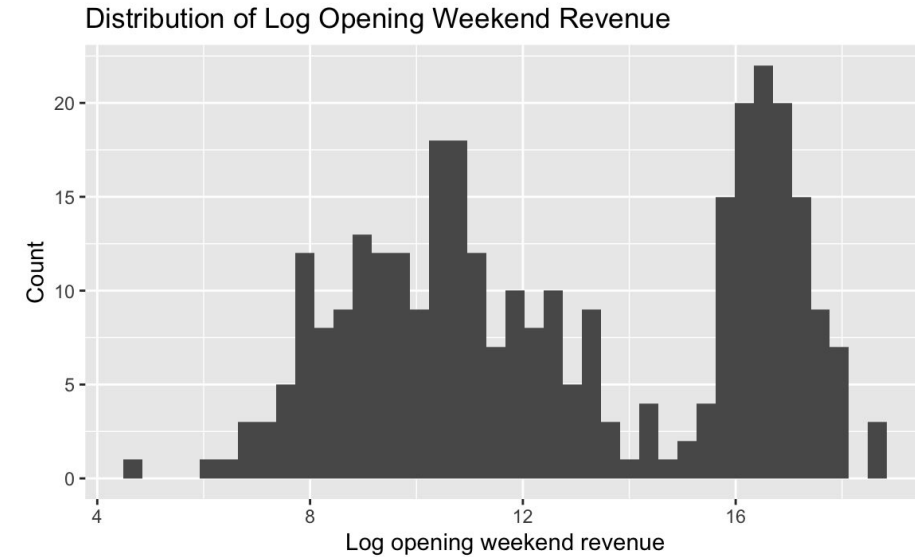
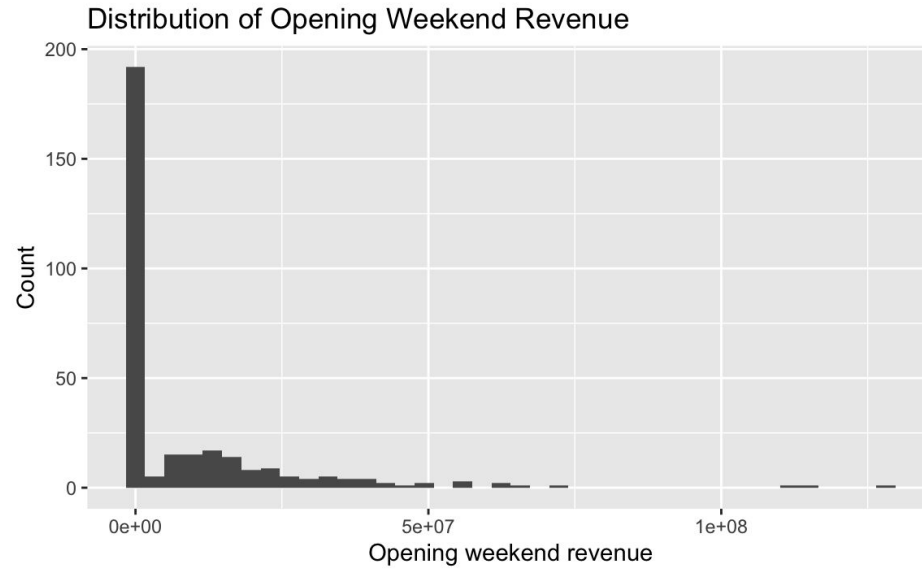
- Single year U.S. sample of 312 movies
- Lower predictive stability for low revenue movies
- Overlap among U,R, and E affects coefficient stability
- Predictive association, not causal evidence

Future Research Directions

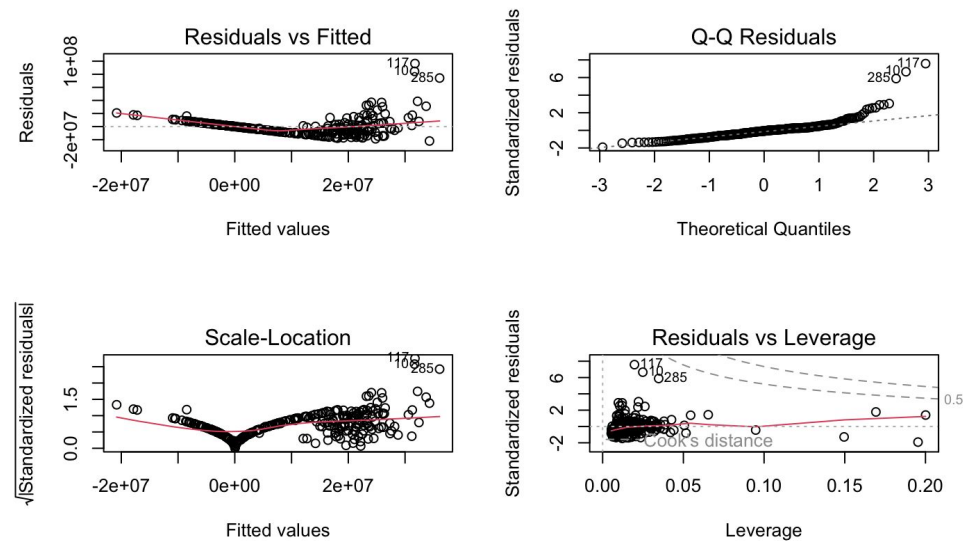
- Validate the model on larger and more diverse datasets
- Use temporal holdout validation to better mimic real forecasting
- Compare different prediction horizons
- Use more pre-release features, such as genre, distributor, budget, trailer activity, and social media signals
- Explore regularized models to handle overlapping editing variables

Q&A?

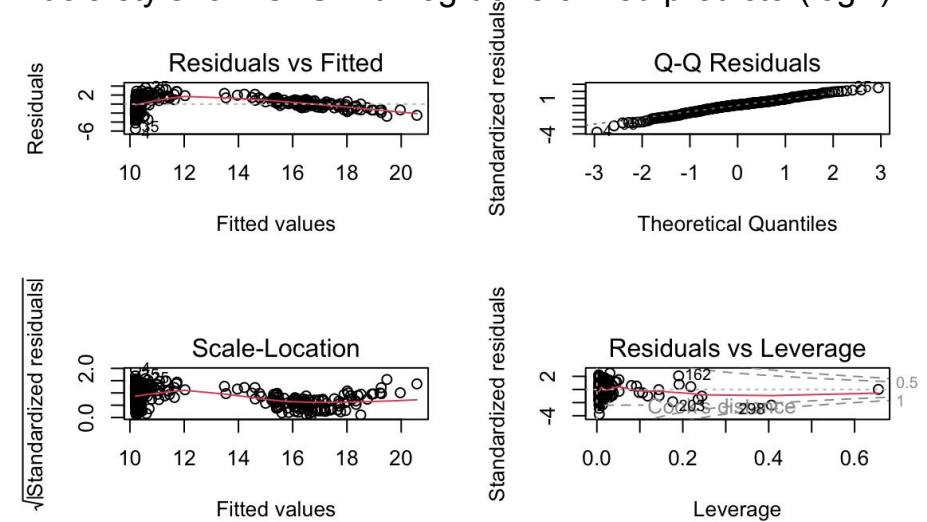
Appendix: Additional Figures



Article-style raw OLS with log-transformed variable(logX)



Article-style raw OLS with log-transformed predictor(logY)



Appendix: Additional Figures

